Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 8 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 8 : SLO â€“ 2 | Queue Implementation Using Array

■ I. Enqueue Operation â€” Insert an Element NUM

• Step 1: IF REAR = MAX - 1
• Write OVERFLOW
• Goto Step 4
• [END OF IF]
• Step 2: IF FRONT = -1 AND REAR = -1
• SET FRONT = REAR = 0
• ELSE
• SET REAR = REAR + 1
• [END OF IF]
• Step 3: SET QUEUE[REAR] = NUM
• Step 4: EXIT

■ II. Dequeue Operation â€” Delete an Element

• Step 1: IF FRONT = -1 OR FRONT > REAR
• Write UNDERFLOW
• ELSE
• SET VAL = QUEUE[FRONT]
• SET FRONT = FRONT + 1
• [END OF IF]
• Step 2: EXIT